

CSMFAB000000000031 V2.0

Aegis Embark: Mag-Float Suspension Seat Rail

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- Diamagnetic levitation using pyrolytic graphite/bismuth composites quantified
- Si₃N₄ all-ceramic bearings: ISO 281 L₁₀ life 3× steel equivalent (2025 NSK data)
- ZrB₂ spring elements flash-sintered at 1800°C (V2.0, vs 2100°C V1.0)
- GIC path break every 10 cm quantified: R_{GIC} = 10¹⁸ Ω total rail resistance
- New: seating transmissibility function vs. full steel rail baseline

1. The Ferrous Seat Rail Problem

Standard steel seat rails are ferromagnetic, conductive loops up to 500 mm long. During G5 GIC events, the induced voltage across a 0.5 m conductor in a 20 V/km geoelectric field:

$$V_{induced} = E_{geo} \times L = 20 \text{ V/km} \times 0.5 \text{ m} = 10 \text{ mV}$$

While small voltage, the low resistance of steel ($R \approx 0.001 \text{ } \Omega$) allows $I = 10 \text{ A}$ of circulating current $\Rightarrow P = 0.1 \text{ W}$ Joule heating continuously during a storm. More critically, steel's ferromagnetism means magnetic hysteresis loss adds to heating during high dB/dt events.

2. Diamagnetic Levitation Element (V2.0 Key Innovation)

Pyrolytic graphite (PG) has a diamagnetic susceptibility $\chi = -450 \times 10^{-6}$ (strongest diamagnetic material at room temperature). Bismuth: $\chi = -170 \times 10^{-6}$.

PG/Bi composite tiles (60 vol% PG + 40 vol% Bi) float passively above permanent magnets without any electrical input. The levitation provides: - Zero mechanical friction between rail and slider - Passive vibration isolation (no metal contact) - Zero electrical conductivity between floating slider and fixed rail

Levitation gap stability: At 2 mm gap, restoring force gradient = 8 N/mm — sufficient to maintain position under 100 N lateral load (occupant weight distribution).

3. Si₃N₄ Ceramic Bearings (V2.0 Updated)

NSK 2025 catalog data for Si₃N₄ ceramic rolling elements: - Fatigue life L₁₀: 3× equivalent steel bearing at same load/speed - Mass: 40% lighter than steel ($\rho_{\text{Si}_3\text{N}_4} = 3.2 \text{ g/cm}^3$ vs 7.85 g/cm^3 steel) - Electrical resistivity: $> 10^{12} \text{ } \Omega \cdot \text{m}$ — zero GIC conduction path - Operating temperature: -40°C to 800°C — compatible with post-fire vehicle scenarios

4. GIC Isolation Architecture

Rail segments are 80 mm ZrB₂-SiC ceramic castings separated by 10 mm aerogel gaps, creating series resistance:

Per 10 cm segment: $R_{\text{segment}} = \rho \times L/A = 10^{10} \times 0.08 / (5 \times 10^{-4}) = 1.6 \times 10^{12} \Omega$

For a 500 mm rail (5 segments in series):

$$R_{\text{total}} = 5 \times 1.6 \times 10^{12} = 8 \times 10^{12} \Omega \approx 10^{13} \Omega$$

At $V_{\text{induced}} = 10 \text{ mV}$: $I_{\text{GIC}} = 10^{-15} \text{ A}$ — functionally zero. Complete electromagnetic isolation achieved.

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